

Yubin Kim

Director of Technology

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Summary

AI/ML leader enthusiastic about building multi-disciplinary teams to solve problems at scale. Experienced leader of machine learning, natural language processing, and search teams in a major healthcare organization working directly with stakeholders across clinical and insurance divisions. Strong research background in information retrieval, NLP, ML with active ties to the research community. Software engineering background, extensive experience in working with large datasets. Knowledgeable in neural NLP methods, classic supervised and unsupervised ML approaches.

Education

- Ph.D. in Computer Science (Dec 2018)
Language Technologies Institute, School of Computer Science
Carnegie Mellon University, Pittsburgh, Pennsylvania
- Bachelor of Software Engineering (Apr 2011)
University of Waterloo, Waterloo, Ontario

Technical Proficiency

- Tools: Java, C++, Ruby, Python, Anaconda, Jupyter notebook, AWS ML ecosystem, Linux, macOS, Windows, git, svn, mercurial
- Information retrieval research areas: distributed search, efficiency, evaluation, learning-to-rank, query analysis, knowledge graphs
- Deep learning NLP approaches including RNNs (e.g. LSTM) and Transformers (e.g. BERT variants)
- Experienced with other supervised and unsupervised methods in ML applied to information retrieval and NLP tasks (e.g. k-means clustering, LDA topic modelling, SVM, tree-based classification, logistic regression)
- Experience in modifying search index and database internals in commercial and research settings

Professional Experience

- Director of Technology at **UPMC Enterprises** (07/2019 - current)
 - Managed team of machine learning engineers, responsible for creating career paths, hiring, performance management, technical mentorship
 - Collaborated with product and stakeholders in both provider and payor divisions to solve problems in healthcare such as claims processing, automated coding, risk stratification, retrospective analysis of intervention effectiveness using ML, NLP, and statistical tools
 - Successfully pitched stakeholder-backed proposals estimating millions in savings to senior leadership
 - Managed relationships with ML/NLP-relevant start-ups, negotiating contracts
 - Reviewed ML/NLP grant proposals for the Pittsburgh Health Data Alliance program
 - Previously led team of product and engineering building Elasticsearch-based search engine over 160+ million clinical documents
- Senior Data Scientist at **UPMC Enterprises** (09/2018 - 07/2019)
 - Developed a prototype of search engine tailored for searching social determinants of health indicators in clinical documents

- Research Assistant at **Carnegie Mellon University** (09/2011 - 12/2018)
 - Advised by Prof. Jamie Callan in the Language Technologies Institute
 - Improved efficiency, accuracy, and stability of selective search, a distributed search architecture that reduces computational cost of web-scale search by 90+%
 - Experiments conducted using data sets of 25–500 million web pages on multi-machine computing clusters
 - Created a learning-to-rank based resource selection algorithm for accurately matching queries to topic-based index shards
- Research Assistant at **Microsoft Research, Redmond** (05/2013 - 08/2013)
 - Integrated crowdsourcing components into information retrieval tasks
 - Mined entity attributes to improve search effectiveness and user experience using crowdsourcing components
- Undergraduate internships at Amazon (2009), Google (2008), Sybase (2007)

Publications

- Carsten Eickhoff, **Yubin Kim**, Ryen W. White. 2020. Overview of the Health Search and Data Mining (HSDM 2020) Workshop. In *Proceedings of the 13th International Conference on Web Search and Data Mining*. 901–902.
- Nithin Haridas, **Yubin Kim**. 2020. Clustering Large-scale Diverse Electronic Medical Records to Aid Annotation for Generic Named Entity Recognition. In *Health Search and Data Mining Workshop*. 43–50.
- Nicholas Ferro, **Yubin Kim**, Mark Sanderson. Using Collection Shards to Study Retrieval Performance Effect Sizes. *ACM Transactions on Information Systems* 37, 3 (2019), 1–40.
- **Yubin Kim**. 2019. Robust Selective Search.
- **Yubin Kim** and Jamie Callan. 2018. Measuring the effectiveness of selective search index partitions without supervision. In *Proceedings of the 4th ACM SIGIR International Conference on the Theory of Information Retrieval*. 91–98.
- Zhuyun Dai, **Yubin Kim**, Jamie Callan. 2017. Learning to Rank Resources. In *Proceedings of the 40th Annual ACM SIGIR Conference*. 837–840.
- **Yubin Kim**, Jamie Callan, Shane Culpepper, Alistair Moffat. Efficient Distributed Selective Search. *Information Retrieval Journal* 20, 3 (2017), 221–252.
- **Yubin Kim**, Jamie Callan, Shane Culpepper, Alistair Moffat. 2016. Load-Balancing in Distributed Selective Search. In *Proceedings of the 39th Annual ACM SIGIR Conference*. 905–908.
- **Yubin Kim**, Jamie Callan, Shane Culpepper, Alistair Moffat. 2016. Does Selective Search Benefit from WAND Optimization?. In *Proceedings of the 38th European Conference on Information Retrieval*. 145–158.
- **Yubin Kim**, Kevyn Collins-Thompson, Jaime Teevan. Using the Crowd to Improve Search Result Ranking and the Search Experience. *ACM Transactions on Intelligent Systems and Technology Special Issue on the Crowd in Intelligent Systems* 7, 4 (2016), 50:1–50:22.
- Zhuyun Dai, **Yubin Kim**, Jamie Callan. 2015. How Random Decisions Affect Selective Distributed Search. In *Proceedings of the 38th Annual ACM SIGIR Conference*. 771–774.
- **Yubin Kim**, Kevyn Collins-Thompson, Jaime Teevan. 2013. Crowdsourcing for Robustness in Web Search. In *Proceedings of the Twenty Second Text REtrieval Conference*. National Institute of Standards and Technology, special publication.
- Jaime Teevan, Kevyn Collins-Thompson, Ryen W. White, Susan T. Dumais, **Yubin Kim**. 2013. Slow Search: Information Retrieval without Time Constraints. In *Proceedings of the 7th Annual Symposium on Human-Computer Interaction and Information Retrieval*. Article 1.
- **Yubin Kim**, Reyyan Yeniterzi, Jamie Callan. 2012. Overcoming Vocabulary Limitations in Twitter Microblogs. In *Proceedings of the Twenty First Text REtrieval Conference*. National Institute of Standards and Technology, special publication.

- George Beskales, Mohamed A. Soliman, Ihab F. Ilyas, Shai Ben-David, **Yubin Kim**. 2010. ProbClean: A probabilistic duplicate detection system. In *IEEE 26th International Conference on Data Engineering*. 1193–1196.

Academic Service

- Organized Workshops
 - Health Search and Data Mining, co-located with the ACM Int'l Conference on Web Search and Data Mining (2020)
- Peer Review
 - Editorial board of Foundations and Trends in Information Retrieval (FnTIR)
 - Program committee of ACM Conference on Information and Knowledge Management (CIKM)
 - Program committee of ACM Int'l Conference on Research and Development in Information Retrieval (SIGIR)
 - Program committee of ACM Int'l Conference on Web Search and Data Mining (WSDM)
 - Program committee of International World Wide Web Conference (TheWebConf)
- Invited Talks and Panels
 - Panelist for "Data for Artificial Intelligence" hosted by *Value for Health CoLAB* (2021)
 - Panelist for "Top of Mind Exchange: AI" hosted by the *Center for Connected Medicine* (2020)
 - Invited talk titled "40,000 Note Types: EMRs in the Real World" presented at the *Artificial Intelligence for Data Discovery and Reuse Symposium* (2020), presented at Brown University (2019)